

GEOMETRY

$$\vec{AB} + \vec{BC} = \vec{AC}$$

It's a geometric construction

length = norm $\rightarrow$ symbol $\| \|$

$\rightarrow$ Addition $\rightarrow$ put a vector after another $\rightarrow$ doesn't need coordinate system

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$$

$\ominus$ change direction -

$\rightarrow$ Scalar Multiplication $\rightarrow$ Multiply the length

$$2 \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

$$k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$$

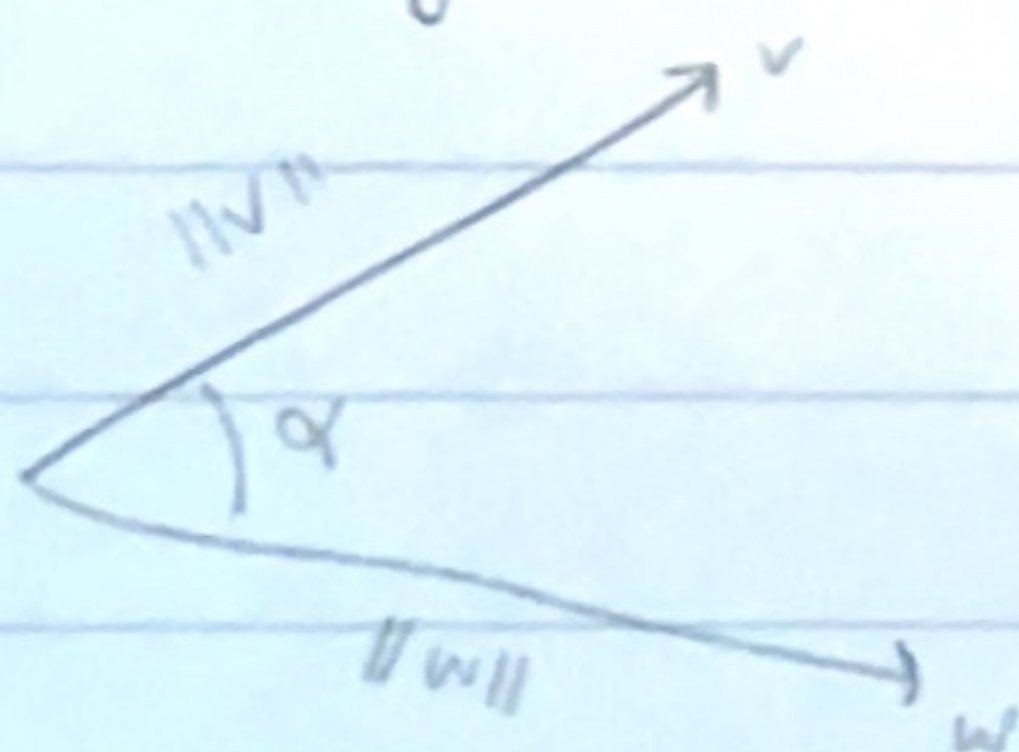
The dot product

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} c \\ d \end{pmatrix} = ac + bd$$

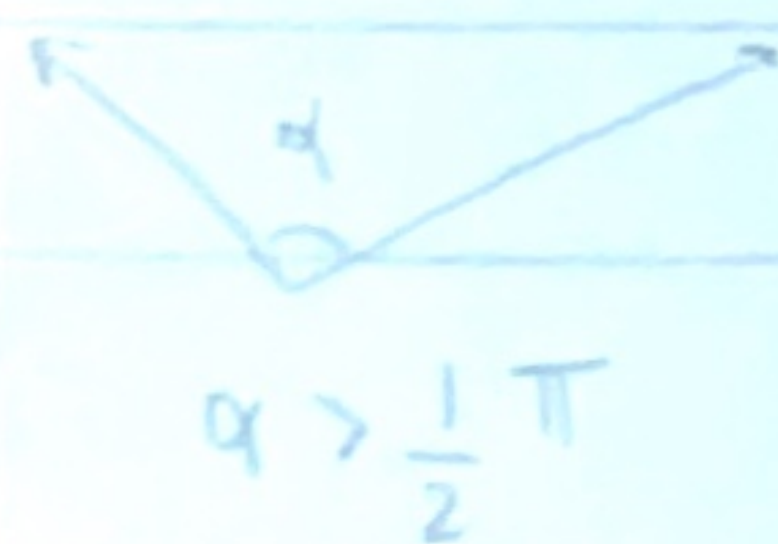
$\leftarrow$ result is a scalar

geometric meaning = $\vec{v} \cdot \vec{w} = \|\vec{v}\| \cdot \|\vec{w}\| \cdot \cos(\alpha) \rightarrow$

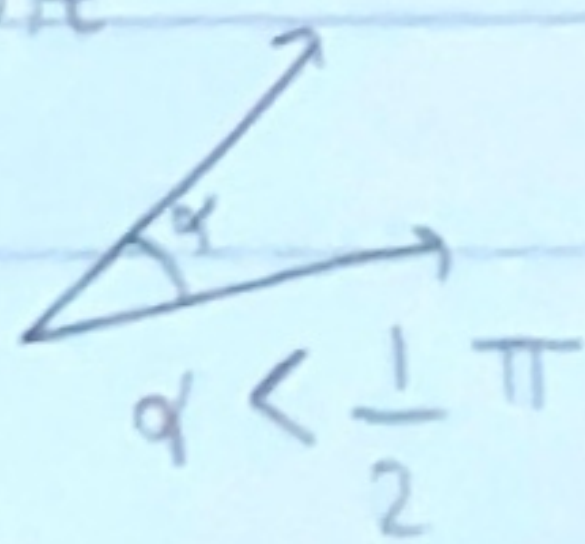
$$\cos(\alpha) = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \cdot \|\vec{w}\|}$$



obtuse



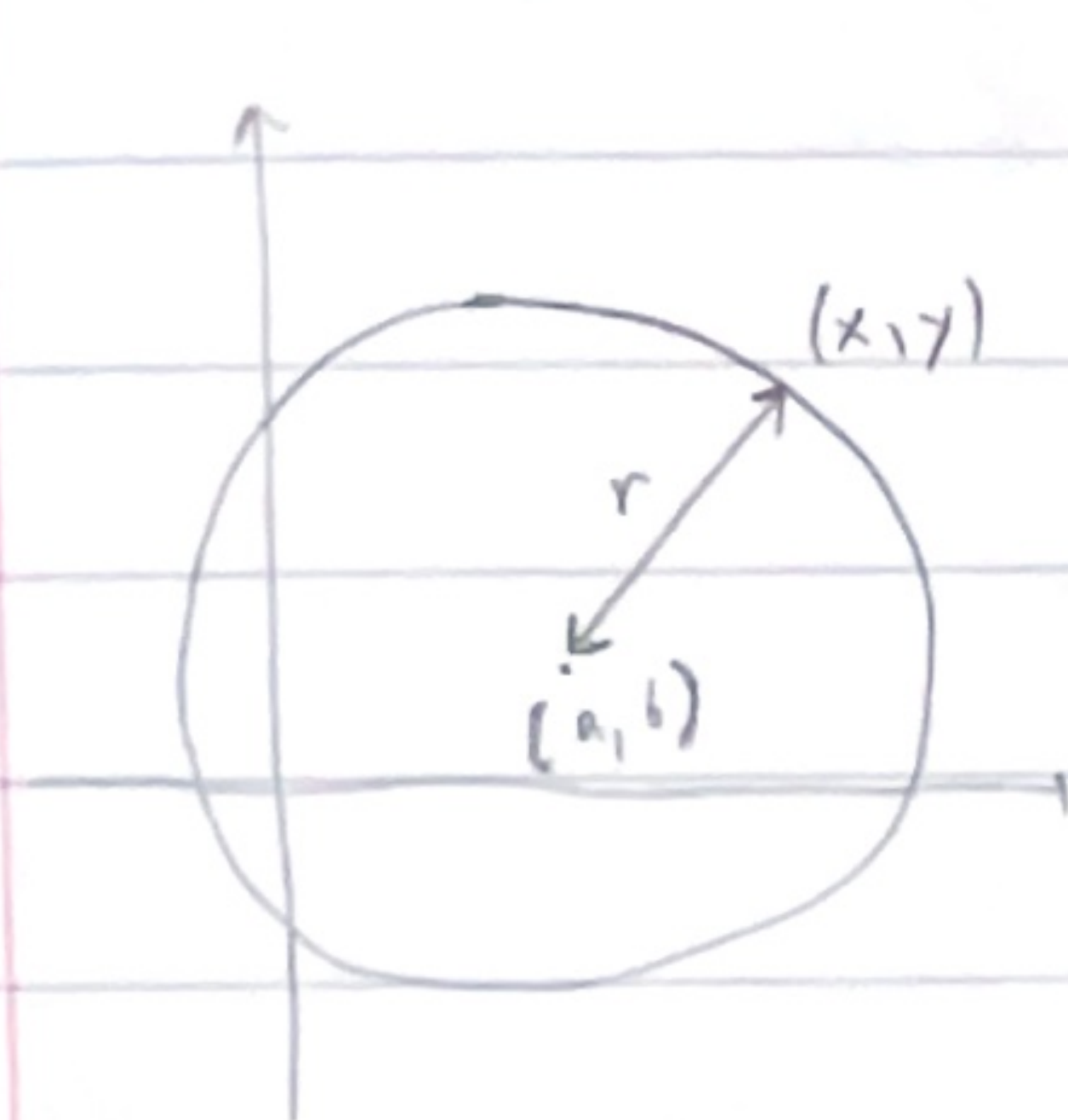
Acute



Right angle

$$\alpha = \frac{1}{2} \pi$$

A CIRCLE



center (a, b)

radius r

$$(x - a)^2 + (y - b)^2 = r^2$$

$$2x^2 - 4x + 2y^2 + 8y + 1 = 0$$

→ Complete the square -

$$\begin{aligned} 2x^2 - 4x &= 2(x - a)^2 + c \\ &= 2x^2 - 4ax + 2a^2 + c \end{aligned}$$

$$2x^2 - 4x = 2(x - 1)^2 - 2$$

$$2y^2 + 8y = 2(y + 2)^2 - 8$$

↓

$$2(x - 1)^2 + 2(y + 2)^2 - 9$$

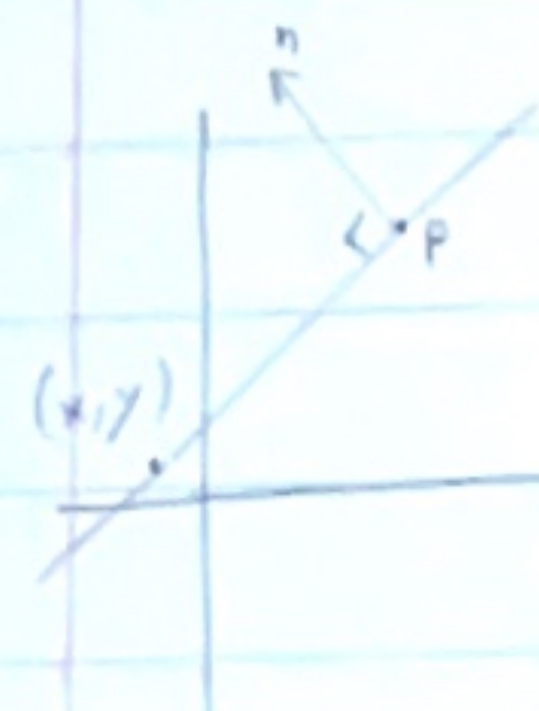
↓

$$(x - 1)^2 + (y + 2)^2 = \frac{9}{2}$$

center $1, -2$

radius $\sqrt{9/2}$ or $\frac{3}{2}\sqrt{2}$

A LINE



(x,y) is on the line if

$$n \perp \begin{pmatrix} x \\ y \end{pmatrix} - P$$

$$n \cdot \left(\begin{pmatrix} x \\ y \end{pmatrix} - P \right) = 0$$

$$n \cdot \begin{pmatrix} x \\ y \end{pmatrix} = n \cdot P$$

$$n = \begin{pmatrix} a \\ b \end{pmatrix} \quad n \cdot P = c$$

↓

to find the equation $ax + by = c$

we need



→ Normal Vector → Swap 2 components and multiply one by -1

direction = from $(4,-2)$ to $(2,1)$ so → $(2,-3)$

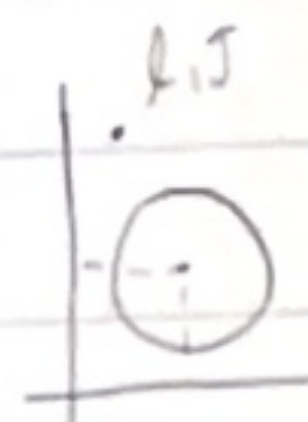
=

$(3,2)$

$$3x + 2y = c$$

→ substituímos por cualquier punto en la línea (2,1)
y obtenemos c

Parametrizations



we can grab a random point (h,j) , substitute in the circle equation and see if that point is on the curve or not

Description of a circle = centre $(4,3)$

radius 2

Implicit description

$$(x-4)^2 + (y-3)^2 = 2^2$$

↓

the path travelled by the mouse

$$x(t) = 4 - 2\cos(t)$$

$$y(t) = 3 + 2\sin(t)$$

to make sure "the mouse only loops the circle once" → restrict values of t to interval $(0, 2\pi)$

Parametrization is an

EXPLICIT description

$$0 \leq t < 2\pi$$

Valid parametrization =

→ All points on curve are covered

→ all parts of curve covered only once (except perhaps single points)

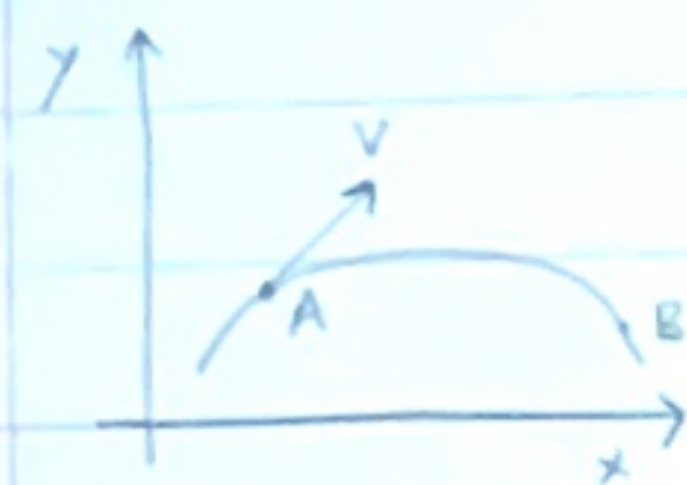
helpful = finding all the points on the curve

velocity and acceleration

velocity = $\vec{v}$

↳ its norm $\|\vec{v}\|$ = speed

find the velocity = curved path



$$\vec{AB} = \begin{pmatrix} x(t + \Delta t) - x(t) \\ y(t + \Delta t) - y(t) \end{pmatrix}$$

Average velocity =

$$V_{av} = \frac{1}{\Delta t} \vec{AB}$$

Velocity at A (time t)

$$v(t) = \lim_{\Delta t \rightarrow 0} V_{av}$$

↓

velocity at time t : $v(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \begin{pmatrix} x(t + \Delta t) - x(t) \\ y(t + \Delta t) - y(t) \end{pmatrix}$

$$\lim_{\Delta t \rightarrow 0} \begin{pmatrix} \frac{x(t + \Delta t) - x(t)}{\Delta t} \\ \frac{y(t + \Delta t) - y(t)}{\Delta t} \end{pmatrix}$$

Meaning if I know the position at time t

↳ I can find

$$= \frac{x'(t)}{y'(t)}$$

the velocity at time t

Velocity at time t : $v(t) = \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix}$

$$\text{Speed at } t = \|\vec{v}(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2}$$

Acceleration $a(t) = v'(t) = \begin{pmatrix} x''(t) \\ y''(t) \end{pmatrix}$



Speed AND direction